DSSI Challenge 2026: Agentic AI for Materials Science

CASC
Center for Applied Scientific Computing

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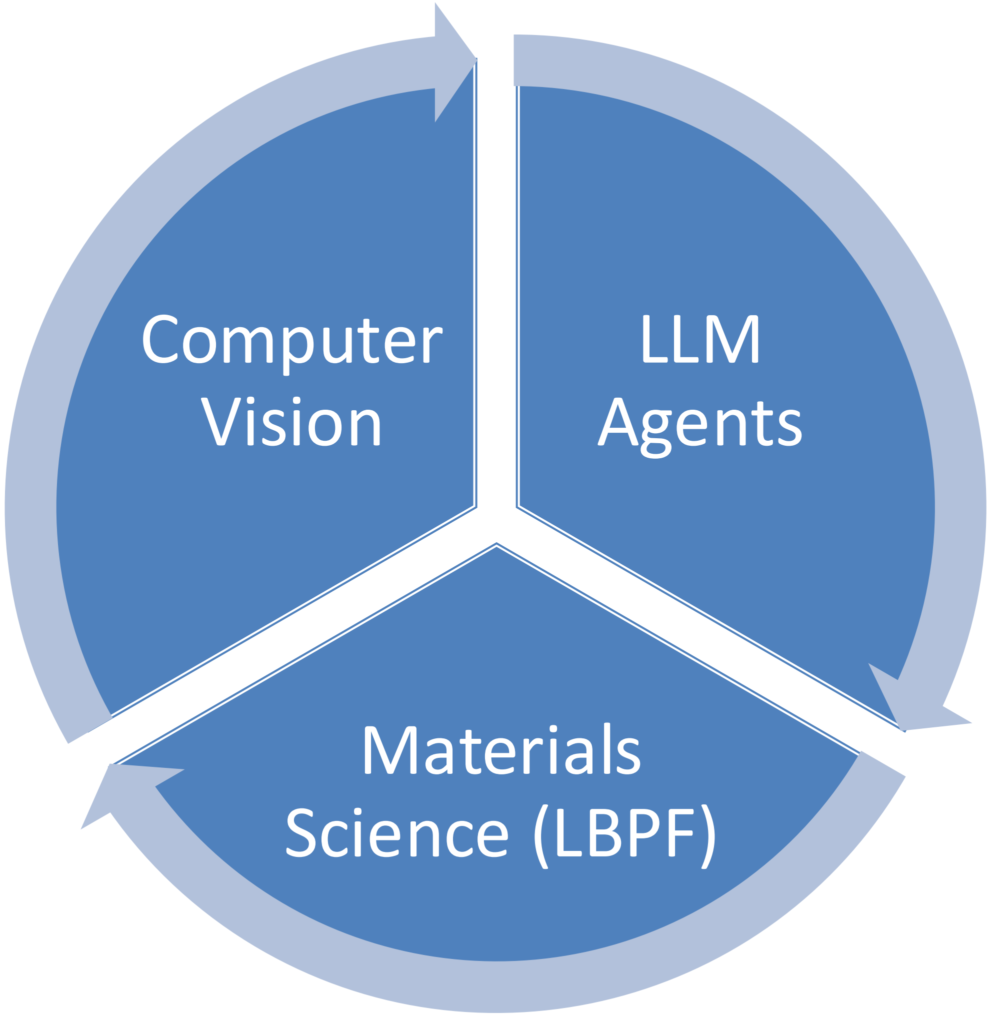
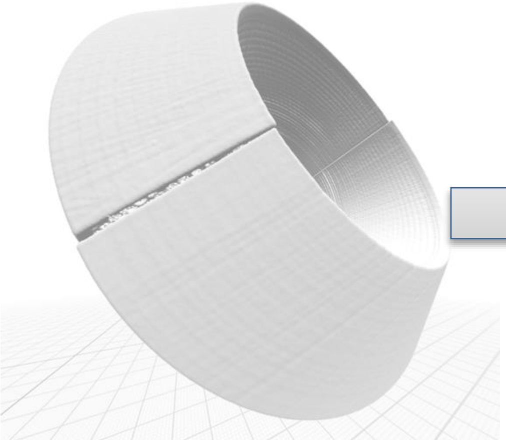
Goal: Build autonomous AI workflows for materials inspection

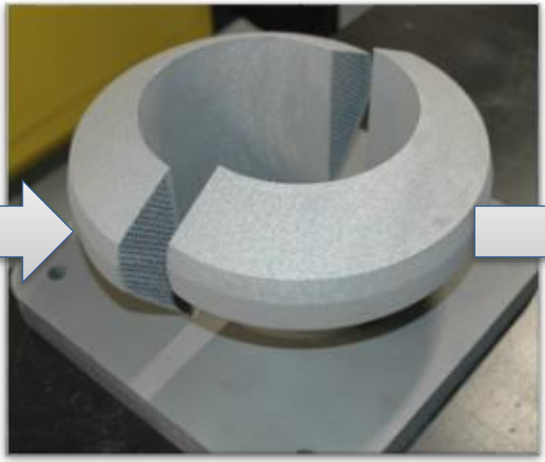


- Build agents that can call tools, inspect outputs, specialize, and evaluate results.
- Work with data from x-ray CT (to see inside a part)
- Apply to defect inspection in additive manufacturing technique, specifically laser-powder bed fusion (LBPF)

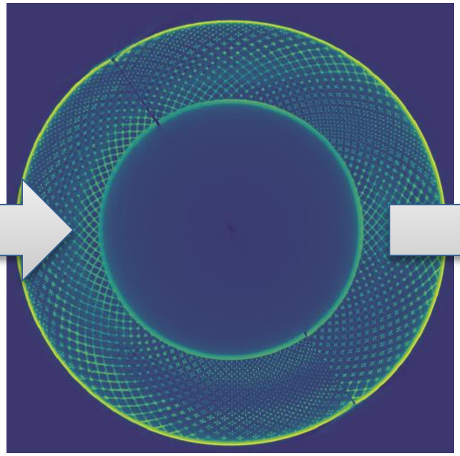
Additive manufacturing & inspection



Design: CAD Model



Print (LBPF)



Scan (x-ray CT)

Analysis

Inspect: Manual

Did we print the right part?

Inspection is the main bottleneck in additive manufacturing

Laser-powder bed fusion (LPBF)

- A metal 3D-printing process where a focused laser selectively melts thin layers of powder to build a dense part, one cross-section at a time.
- Why it matters: complex channels, lightweight lattices, rapid iteration, and high-performance metal parts.
- Typical metals: stainless steels, titanium alloys, nickel superalloys, aluminum alloys.

1 **Spread powder**

Recoater lays down a thin metal layer.

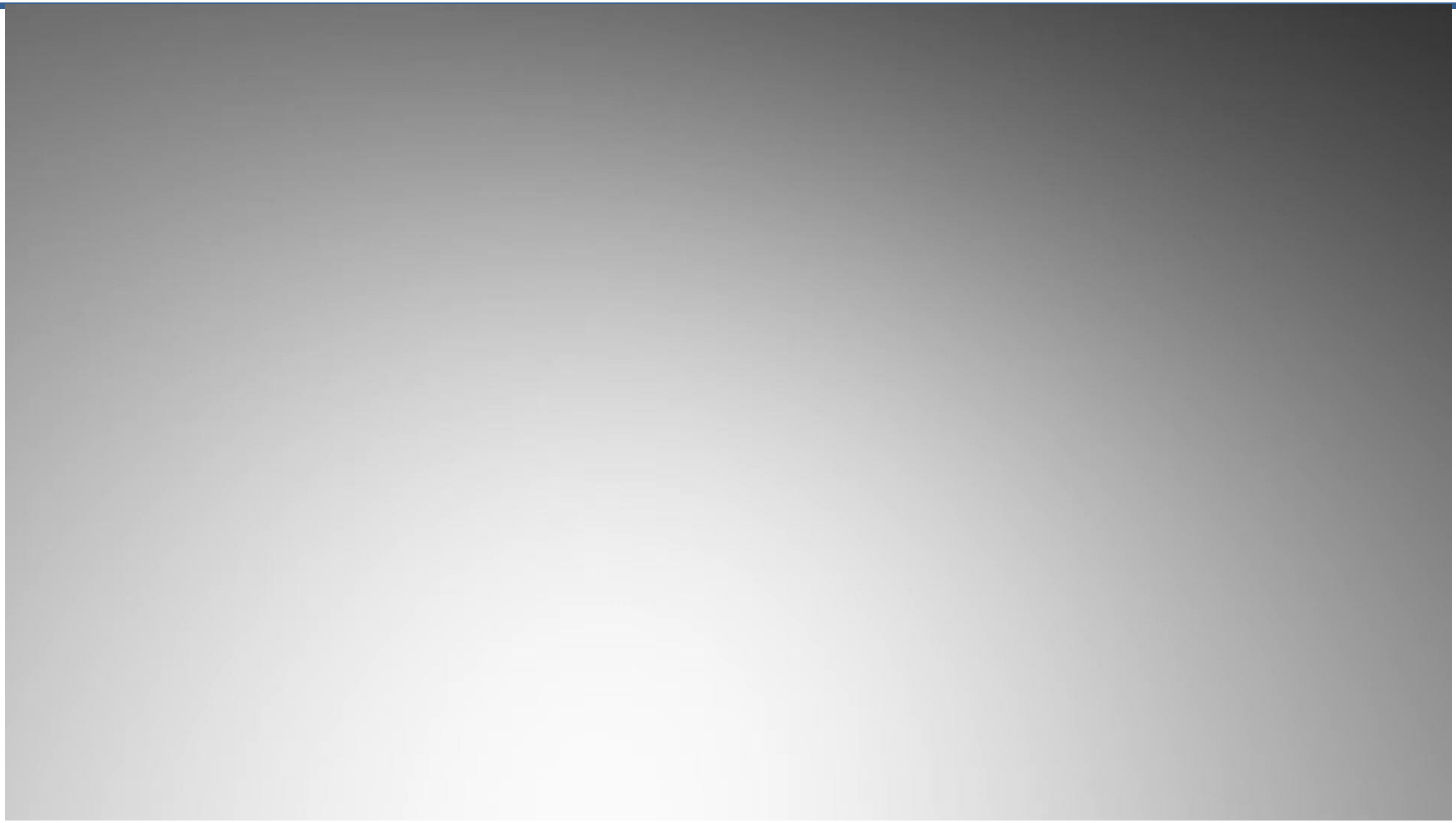
2 **Melt scan path**

Laser fuses the layer cross-section.

3 **Repeat and finish**

Platform lowers; unfused powder is removed.

Laser-powder bed fusion (LPBF)

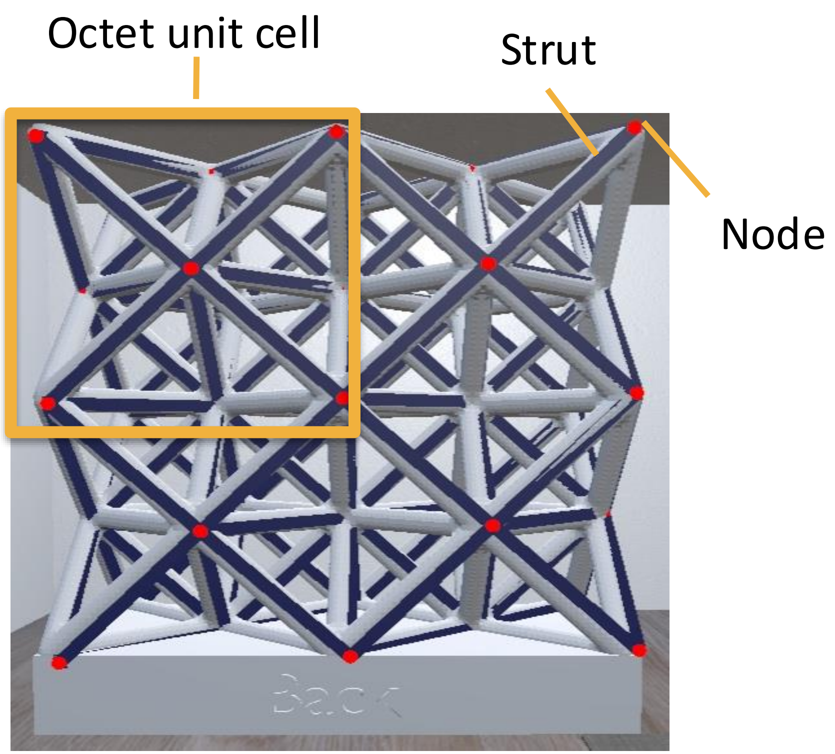


Youtube: What Is Laser Powder Bed Fusion?
AM Media | The Cool Parts Show

Octet Unit Cell & Lattice Structures

- In LPBF, **lattice structures** are engineered **internal geometries** made by repeating small "unit cells" throughout a metal part.
- **An octet unit cell is a strut-based lattice cell** made from interconnected diagonal beams; when repeated, it forms an **octet-truss lattice** with high stiffness and strength at low density.

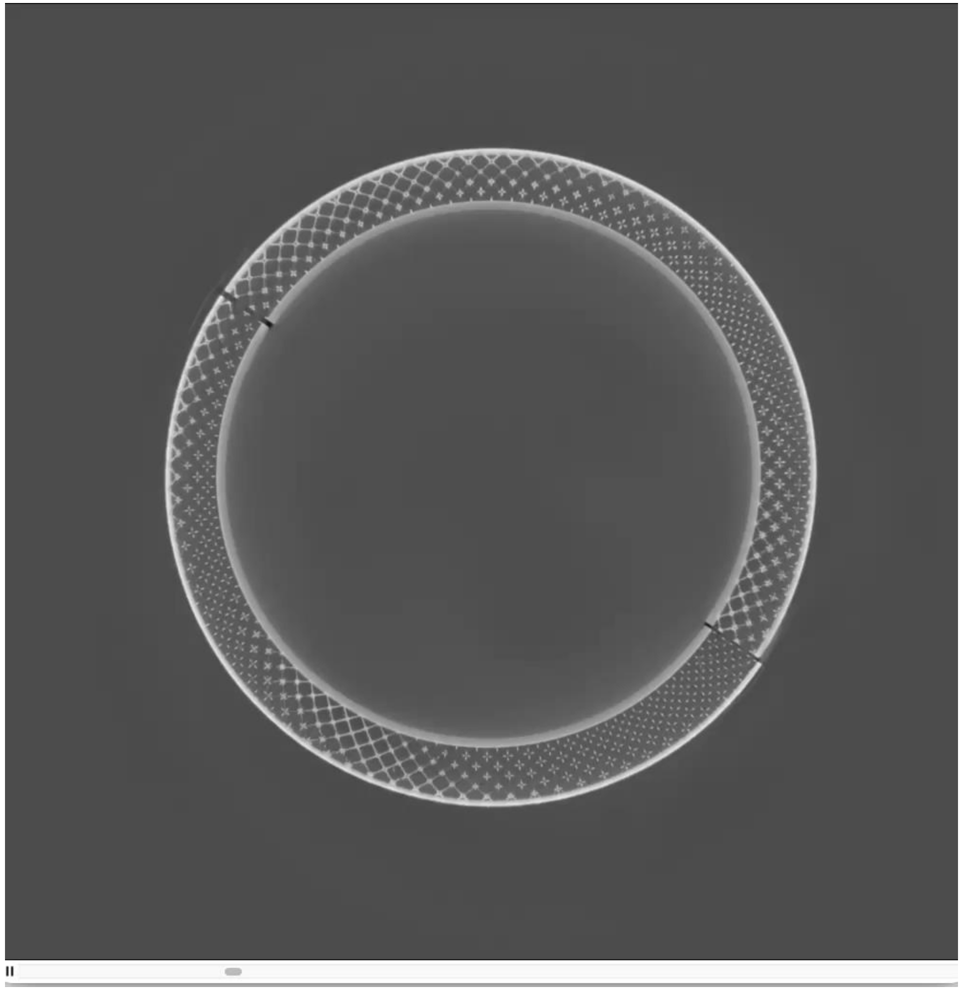
Applications: lightweight, energy absorption, stiffness-to-weight ratio, porous material



2x2x2 Octet Lattice

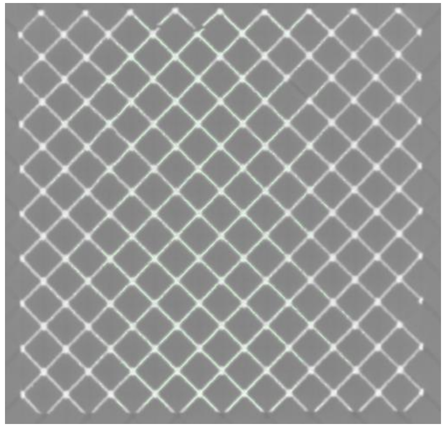
X-ray Computed Tomograph to reveal internal details

- Non-destructive imaging method that uses X-rays to capture many 2D projections of an object from different angles.
- Algorithms reconstruct the projections into a 3D volume
- CT scanning lets engineers “see inside” LPBF parts and quantify internal defects, making it a critical tool for validating complex additively manufactured components.
- Resulting datasets are volumes (3D scalar fields with grayscale density values)

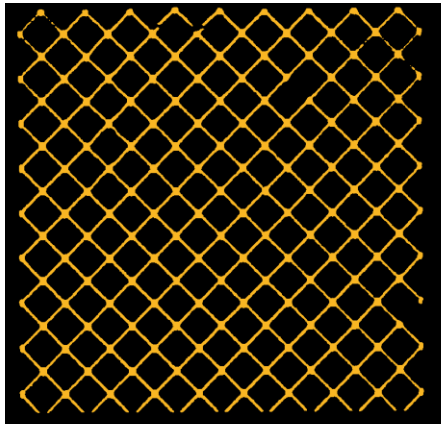


LPBF enables intricate geometries, but inspection becomes the main bottleneck

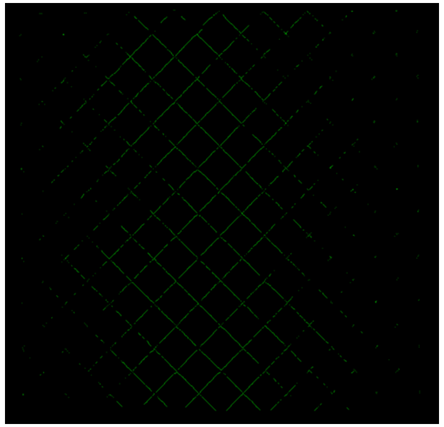
- Missing struts can break load paths.
- Segmentation and skeletonization turn density into structure an agent can reason about.



Raw CT slice
Grayscale density values



Segmentation mask
Material separated from background



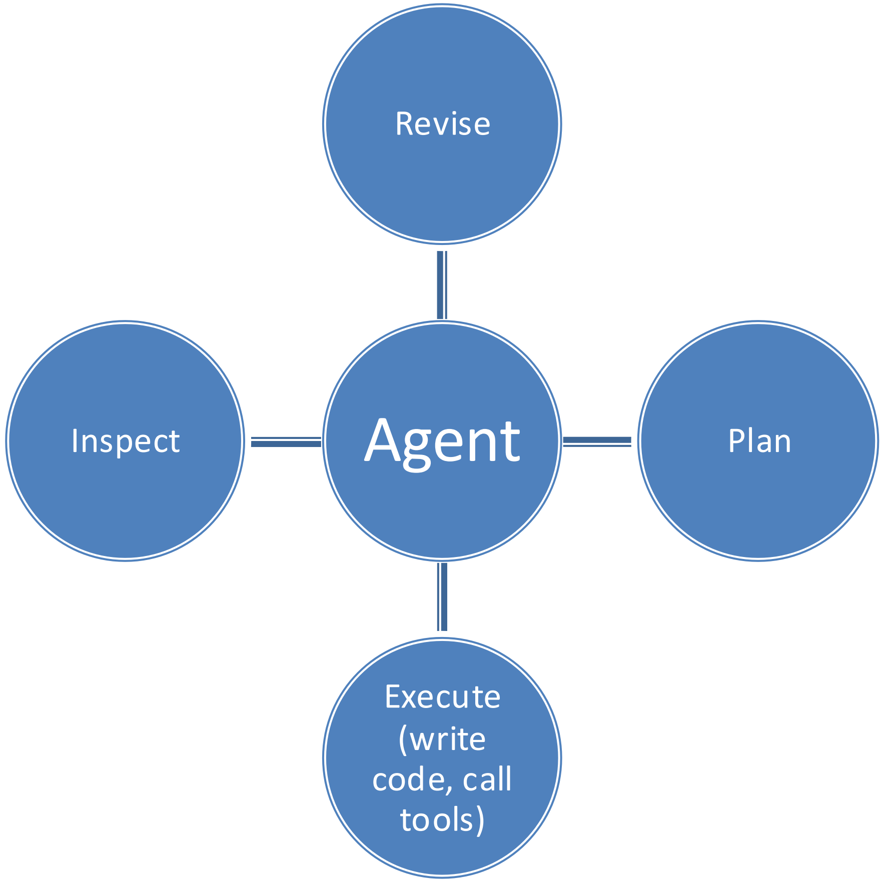
Extracted skeleton
Centerlines for struts, nodes, and connectivity

What’s the role of AI in materials science

- Materials science is becoming **data-rich and workflow-heavy**
- AI is moving from “predict one thing” / “chatbot” to “do a whole workflow”
- Manual workflows become the bottleneck, and algorithm development is too rigid and not scalable

Agents orchestrate scientific workflows, not single predictions

- Agents are useful when progress depends on planning, tool use, inspection, and revision.
  - Plan the next experiment/analysis steps
  - Write & call code, local tools, and analysis routines from existing software.
  - Inspect artifacts: slices, masks, reports, logs, failures
  - Revise parameters or code until the goal is achieved



Why this DSSI challenge is a good training ground

- The CT lattice task exposes students to the full stack of agentic materials workflows.

MCP tools
Expose reliable Python functions for segmentation, slicing, and skeletonization.

Skills
Package domain instructions for NDE reports so the agent writes like a materials scientist.

Subagents
Give specialized workers bounded loops for optimization and traceable outputs.

LLM evals
Judge result images against ground truth with a rubric and JSON score.

Students learn to build systems that can do the science: read data, run tools, inspect outputs, improve parameters, and produce defensible reports.

Code Agent as the Harness

- Codex can coordinate MCP tools, project skills, subagents, scripts, and evaluations from one coding-centered interface.

- **Bridges tools and missing capabilities**
  - When a tool exists, Codex can call it.
  - When a tool does not exist yet, Codex can write the missing code, test it, and integrate it into the workflow.

- Supports self-feedback and iteration
  - Codex can inspect outputs, identify failures, revise code or parameters, and rerun the workflow.
  - This enables closed-loop improvement for tasks like CT segmentation, visualization, and defect detection.

- Codex is not just a code agent; it is a flexible agent harness that can use tools, create tools, evaluate results, and adapt workflows for materials science.

Now let's get started with agentic AI for materials science

- Fork it on Github: https://github.com/mhaichao/llnl_data_science_challenge_2026